

Assignment: Word Importance Explorer (TF-IDF)

Date: 24/03/2026

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Objective

To analyze multiple text documents and identify important keywords using the TF-IDF technique.

Documents

Machine learning is amazing and powerful
Artificial intelligence and machine learning are related
Data science uses machine learning techniques
Deep learning is a part of artificial intelligence
Python is widely used in data science and AI

Methodology (TF-IDF)

Convert text into numerical form using TF-IDF

TF (Term Frequency): Frequency of a word in a document

IDF (Inverse Document Frequency): Importance of a word across all documents

$TF-IDF = TF \times IDF$

Higher score $\rightarrow$ More important word

Python Code

```
from sklearn.feature_extraction.text import TfidfVectorizer
documents = [
```

```
    "Machine learning is amazing and powerful",
    "Artificial intelligence and machine learning are related",
    "Data science uses machine learning techniques",
    "Deep learning is a part of artificial intelligence",
    "Python is widely used in data science and AI"
]
```

```
# Create TF-IDF model
```

```
vectorizer = TfidfVectorizer()
```

```
tfidf_matrix = vectorizer.fit_transform(documents)
```

```
# Get feature names
```

```
features = vectorizer.get_feature_names_out()
```

```
# Convert matrix to array
```

```
tfidf_array = tfidf_matrix.toarray()
```

```
# Print top 3 keywords for each document
```

```
for i in range(len(documents)):
```

```
    print(f"\nDocument {i+1}:")
```

```
    scores = tfidf_array[i]
```

```
    top_indices = scores.argsort()[-3:]
```

```
    for idx in top_indices:
```

```
        print(features[idx], ":", round(scores[idx], 2))
```

Top Keywords (Result)

Doc 1: amazing, powerful, machine

Doc 2: artificial, intelligence, related

Doc 3: data, science, techniques

Doc 4: deep, artificial, learning

Doc 5: python, data, science

Explanation

TF-IDF assigns higher importance to words that are frequent in one document but not common in all documents. This helps in extracting meaningful keywords while ignoring common words.

Conclusion

TF-IDF is an effective method to find important words in text data. It is widely used in search engines, document analysis, and natural language processing.